

git --version

git init

git add .

git commit -m "weather app push"

git push

git remote add origin <https://github.com/hemamku/aim-demo>

git branch -M main

git push -u origin main

git add .gitignore

git commit -m "add gitignore"

git push

git pull --rebase origin main

git commit -m "add gitignore"

git push

git checkout -b develop

git branch

git checkout main

docker build -t weather-app:1.0 .

docker build -t weather-app:1.0 .

docker images

docker run -d --name weather-app -p 5000:5000 weather-app:1.0

docker ps

docker build -t weather-app:1.0 .

docker images

docker run -d --name weather-app -p 5000:5000 weather-app:1.0

docker ps

docker run -d --name weather-app -p 5000:5000 weather-app:1.0

docker run -d --name weather-app -p 5000:5000 weather-app:1.0

docker ps

git add .

git commit -m "weather app"

git push -u origin feature/improve-weather-ui

docker build -t weather-app:1.0 .

docker images

docker run -d --name weather-app -p 5000:5000 weather-app:1.0

docker rm -f weather-app

docker run -d --name weather-app -p 5000:5000 weather-app:1.0

docker ps

<http://localhost:5000/>

open docker desktop and Jenkins(<http://localhost:8080>)

Jenkins user name: admin

pwd cmd:

PS C:\Users\HEMA>

docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword

create new item as pipeline:

Pipeline

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|— Definition

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|— Pipeline script from SCM

|

|— SCM

|

|— Git

|

|— Repository URL

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|— <https://github.com/hemamku/aim-demo.git>

|

└─ Branch Specifier

| └─ */main

|

└─ Script Path

└─ Jenkinsfile

Click Buildnow option once item/job is created.. It should successfully run and finish the build..

Next install minikube: In powershell

winget install Kubernetes.minikube

minikube version

kubectrl version --client

Start the minikube :

minikube start --driver=docker

result should be: * Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default

minikube status

kubectrl config current-context

kubectrl get nodes

Phase-1:

Docker → Minikube → Kubernetes

In powershell:

docker images weather-app

minikube image load weather-app:latest

minikube image ls

SO now... :

Docker Desktop

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| minikube image load

↓

Minikube's image environment

Create a new folder k8s under weather-app and place deployment.yaml and service.yaml file and then run this in powershell in the weather-app directory:

```
kubectl apply -f k8s/deployment.yaml
```

```
kubectl apply -f k8s/service.yaml
```

```
kubectl get pods
```

```
kubectl get deployment
```

```
kubectl get service
```

```
minikube service weather-app-service --url
```

```
minikube dashboard
```

Git

```
winget install --id Git.Git -e --source winget
```

Java 21 for Jenkins

```
winget install --id EclipseAdoptium.Temurin.21.JDK -e
```

Docker Desktop

```
winget install --id Docker.DockerDesktop -e
```

Minikube

```
winget install Kubernetes.minikube
```

kubectl

```
winget install Kubernetes.kubectl
```